

Password Door Lock

FSM-Based Digital Lock in Verilog

Digital Logic Design — Project Report

1. Overview

This project implements a synchronous hexadecimal password door lock as a Mealy finite state machine (FSM), split across three Verilog modules. On power-up the system has no password saved. A 32-bit password is programmed once, after which the system continuously compares any new input against the saved password. Four wrong attempts in a row trip an alarm and freeze the system completely; only a full reset can recover it, and after that reset the password must be programmed again.

2. System States

The design uses three states:

- INITIAL — no password saved yet. Waits for `i_set_en` to program the first password, then moves to VERIFY.
- VERIFY — normal operating state. Compares every input against the saved password: a match grants access and resets the trial counter, a mismatch increments the trial counter.
- FREEZE — reached after 4 wrong attempts in VERIFY. Alarm stays on, access stays denied, and no input is accepted until a reset.

3. Module Breakdown

File	Role
password_fsm.v	The Mealy FSM: state register, trial counter, and all control outputs (<code>o_access</code> , <code>o_alarm</code> , <code>o_frozen</code> , <code>o_load</code> , <code>o_trials</code> , <code>o_state</code>)
password_reg.v	Stores the currently saved 32-bit password, loaded on <code>i_load</code> from the FSM
password_top.v	Top-level module wiring <code>password_fsm</code> and <code>password_reg</code> together
testbench_top.v	Testbench exercising password programming, 4 wrong attempts, freeze, and reset recovery

4. Top-Level Interface

Signal	Direction	Description
<code>i_input[31:0]</code>	Input	32-bit value entered by the user
<code>i_set_en</code>	Input	Programs the password once, from the INITIAL state
<code>i_reset</code>	Input	Clears the FSM, the trial counter, and the saved password
<code>i_clk</code>	Input	System clock, all logic is synchronous
<code>o_access</code>	Output	High when the entered password matches the saved one
<code>o_alarm</code>	Output	High after 4 consecutive wrong attempts
<code>o_frozen</code>	Output	High while the system is locked in FREEZE
<code>o_trials[2:0]</code>	Output	Count of consecutive wrong attempts, 0 to 4
<code>o_state[1:0]</code>	Output	Current FSM state (0=INITIAL, 1=VERIFY, 2=FREEZE)

5. Simulation and Verification

The design was verified in QuestaSim by exercising password programming, four consecutive wrong attempts, the freeze condition, a full reset, and recovery. The waveform below shows the resulting signal transitions.

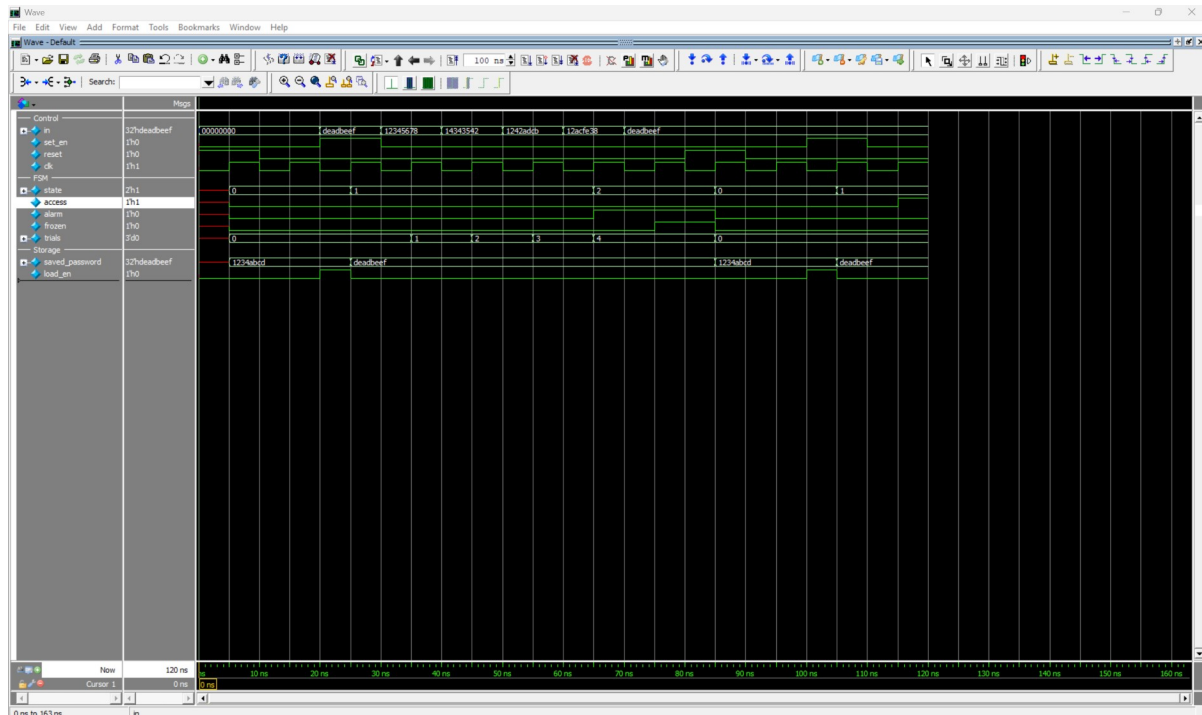


Figure 1 — QuestaSim waveform of the full test scenario

5.1 Observed Results

Time	Event	Observed result
0-10 ns	Power-up reset	state=INITIAL, all outputs and the trial counter are 0
20-25 ns	Password programmed (deadbeef)	state moves to VERIFY, saved_password updates to deadbeef, access=1
30-60 ns	4 wrong attempts	trials increments 1,2,3,4 with each wrong value, access stays 0
65 ns	4th wrong attempt	alarm=1 and state moves to FREEZE
70-75 ns	Correct password entered while frozen	access stays 0, frozen=1 confirms the lock holds
80-90 ns	Reset applied	state returns to INITIAL, saved_password resets to the default value
100-115 ns	Password reprogrammed and re-entered	state moves to VERIFY, access=1 confirms recovery works

6. Conclusion

The waveform confirms the FSM behaves as designed: the password is only accepted while in VERIFY, the trial counter tracks wrong attempts accurately, the system freezes irreversibly after the fourth wrong attempt until reset, and a reset correctly returns the system to INITIAL with the saved password cleared. The one-cycle separation visible between alarm and frozen reflects the registered Mealy outputs and is expected behavior, not a fault.